

Seungwan Seo

Ph.D., in Industrial Engineering

Contact Information

Robot Learning Research Scientist

CarbonSix
331, Gangnam-daero, Seocho-gu, Seoul, Republic of Korea

E-mail: lepoeme20@gmail.com

Github: <https://github.com/lepoeme20>

linkedin: <https://www.linkedin.com/in/lepoeme>

Experience

Robot Learning Research Scientist

CarbonSix

April 2026 -- Present

- Develop VLA/foundation models for SigmaKit, from training to deployment
- Build, curate, and augment robot datasets
- Improve sim-to-real transfer for robust real-world robot performance

Lead Vision Algorithm Engineer

GENGENAI

May 2024 -- April 2026

- Led the research and development of large-scale synthetic image generation pipelines for training and evaluating vision models
- Designed data generation frameworks for non-optical sensors (e.g., SAR and infrared), addressing extreme data scarcity scenarios
- Managed the end-to-end technical execution of client-driven synthetic data projects, from problem formulation to pipeline development

Senior Researcher

AI Vision Cell, Hana Institute of Technology

February 2023 -- May 2024

- Conducted applied research on document understanding and OCR-based classification under real-world noisy conditions
- Developed generative models for virtual human synthesis, including facial expression and lip-motion generation from limited data
- Delivered a real-time virtual human prototype and demonstrated its feasibility at Korea Fintech Week 2023

Education

Korea University

Ph.D. in Industrial Management

Seoul
2017 - 2023

Seoul Tech

Bachelor of Science in Information Technology and Management

Seoul
2011 - 2017

Selected Projects (Industry)

@ CARBONSIX

2026.04 – CURRENT

Develop CarbonSix's proprietary robot foundation model

CarbonSix, Research

2026.08 -- Present

- Train the model from scratch rather than adapting existing open-source WAMs or VLAs

Enhanced ML capabilities in CarbonSix's robot learning system

2026.04 -- Present

CarbonSix, Research

- Introduced Diffusion and Flow Matching for action generation
- Implemented real-time action chunking for smoother robot motion without performance degradation, reducing jerk by over 70%
- Introduced latent-space feature prediction to improve model precision, achieving over a 7 percentage-point accuracy gain on real-world tasks

@ GENGENAI

2024.05 – 2026.04

High-altitude Narrow-FOV Synthetic Image Generation

2025.05 -- 2026.04

Korea Aerospace Industries (Client)

- Generated synthetic imagery simulating observations from altitudes up to 9,000 m (30,000 ft) under extremely narrow field-of-view constraints
- Generated full scenes containing TEL and SAM assets using conditional T2I models, with precise control over viewing geometry and object scale
- Enabled configurable camera tilt (pitch) angles ranging from 15° to 90° to support diverse observation scenarios
- Modeled diverse environmental conditions (e.g., snow, rain, fog) to support data generation across varied weather scenarios

Drone-view Synthetic Data Generation via Conditional T2I Models

2025.01 -- 2025.05

Korea Aerospace Industries (Client)

- Generated drone-view image datasets using conditional text-to-image models, leveraging open datasets such as VisDrone
- Synthesized drone-view scenes with precise spatial control over the placement and type of TEL assets
- Achieved a 7 p.p. improvement in object detection performance (mAP@50) on real drone-captured test images when training exclusively on synthetic data, compared to models trained on open datasets
- Reduced generation latency by replacing inpainting-based pipelines with condition-controlled T2I generation
- Trained detection models capable of identifying TEL decoys with up to 90% detection accuracy (mAP@50)

Satellite-view Synthetic Image Generation for TEL Detection

2024.10 -- 2024.12

Korea Aerospace Industries (Client)

- Generated nadir-view (top-down) satellite imagery with controllable placement of specific TEL variants
- Designed scene-consistent synthetic data pipelines to replicate realistic satellite observation conditions
- Produced fully labeled datasets optimized for object detection under limited availability of real satellite imagery

Core Research on Synthetic Data Generation for Autonomous Driving

2024.05 - 2024.10

GENGENAI, Foundational Research

- Developed scene-aware inpainting techniques to insert vehicles, cyclists, and pedestrians at controllable locations within autonomous driving scenes
- Mitigated severe class imbalance in object detection datasets by selectively synthesizing under-represented object categories
- Validated the effectiveness of the synthetic data through a 16% improvement in object detection performance (mAP@50) compared to training on real-world datasets alone

@HANA INSTITUTE OF TECHNOLOGY

2023.02 – 2024.05

Virtual Human Generation from Limited Visual Data

2023.04 - 2024.05

Hana Institute of Technology, Foundational Research

- Developed generative pipelines for virtual human synthesis using diffusion and StyleGAN-based models
- Generated identity-consistent virtual humans from limited visual inputs, including single-image scenarios
- Implemented facial attribute manipulation and lip motion generation conditioned on user-provided scripts
- Built an end-to-end system for producing talking-head videos, enabling scalable content generation for VTM applications

OCR-based Document Classification for Financial Applications

2023.02 - 2023.05

Hana Bank

- Designed an OCR-based document understanding system for foreign exchange and financial documents
- Addressed classification challenges arising from handwritten text, noisy scans, and heterogeneous document layouts
- Classified document types into fine-grained categories and extracted key phrases to support downstream financial processing

Selected Projects (Doctoral Research)

Synthetic Data Generation for Unseen Ocean Environments

2019.09 - 2020.05

Daewoo Shipbuilding & Marine Engineering

- Generated synthetic ocean imagery using GANs to improve classification performance on unseen data

Ocean Wave Property Prediction from Imagery

2020.12 - 2021.10

Daewoo Shipbuilding & Marine Engineering

- Developed deep learning-based models to predict wave height, direction, and period from ocean wave images

DeepFake Video Detection

2019.03 - 2021.02

National Research Foundation of Korea

- Developed DeepFake detection models using classification and anomaly detection techniques

Sentence Similarity Estimation via Image Association

2018.06 - 2019.02

NCSOFT

- Designed multimodal models to estimate sentence similarity through image-based associative representations

System Anomaly Detection Using Deep Learning

2018.03 - 2019.03

Agency for Defense Development

- Detected abnormal system states using RNN-based autoencoder models

Publications

Exploring the Differences in Adversarial Robustness Between ViT- and CNN-Based Models Using Novel Metrics

2023.10

Seungwan Seo, Jaehyuk Heo, Pilsung Kang*

Computer Vision and Image Understanding, IF: 4.886, Q1 Journal

[Original paper link](#)

Evaluating and improving adversarial robustness of CNNs on image classification

2023.02

Seungwan Seo

Korea University

[Original paper link](#)

Cost-free Adversarial Defense: Distance-based optimization for model robustness without adversarial training

2023.01

Seungwan Seo, Yunseung Lee, Pilsung Kang*

Computer Vision and Image Understanding, IF: 4.886, Q1 Journal

[Original paper link](#)

Intrusion Detection based on Sequential Information preserving Log Embedding Methods and Anomaly Detection Algorithms

2021.04

Czangyeob Kim, Myeongjun Jang, Seungwan Seo, Kyeongchan Park, Pilsung Kang*

IEEE Access, IF: 4.34, Q1 Journal

[Original paper link](#)

Unusual customer response identification and visualization based on text mining and anomaly detection Seungwan Seo, Deokseong Seo, Myeongjun Jang, Jaeyun Jeong, Pilsung Kang* Expert Systems with Applications, Volume 144, April 2020, 113111, IF: 9.60, Q1 Journal Original paper link	2020.04
Comparative Study of Deep Learning-Based Sentiment Classification Seungwan Seo, Czangyeob Kim, Haedong Kim, Kyoung Hyun Mo, Pilsung Kang* IEEE Access, Volume 8, January 2020, Pages 6861 - 6875, IF: 4.34, Q1 Journal Original paper link	2020.01
Recurrent neural network-based semantic variational autoencoder for Sequence-to-sequence learning Myeongjun Jang, Seungwan Seo, Pilsung Kang* Information Sciences, Volume 490, July 2019, Pages 59-73, IF: 6.795, Q1 Journal Original paper link	2019.07
Sentiment classification with word localization based on weakly supervised learning with a convolutional neural network Gichang Lee, Jaeyun Jeong, Seungwan Seo, CzangYeob Kim, Pilsung Kang* Knowledge-Based Systems Volume 152, 15 July 2018, Pages 70-82, IF: 8.038, Q1 Journal Original paper link	2018.07

Conference

An Active Reference Reset Method Adapting Distribution Shift for Robust System Anomaly Detection Seungwan Seo, Heejeong Choi, Pilsung Kang* 2022 13th International Conference on Information and Communication Technology Convergence (ICTC) Original paper link	2022.10
Cost-free Adversarial Defense: Distance-based optimization for model robustness without adversarial training Korean Institute of Industrial Engineers (Spring), Jeju Convention Center, Jeju Seungwan Seo, Yunseng Lee, Pilsung Kang*	2021.06
Intrusion detection based on sequential information preserving log embedding methods and anomaly detection algorithms INFORMS Annual Meeting, Seattle, WA Seungwan Seo, Pilsung Kang*	2019.11
Measuring Sentence Similarity based on Image Association and Siamese Network Korean Institute of Industrial Engineers (Fall), Seoul National University, Seoul Seungwan Seo, Minsung Jeong, Heejeong Choi, Pilsung Kang*	2019.11
Identifying and visualizing uncommon customer response on machine learning INFORMS Annual Meeting, Phoenix, Arizona Seungwan Seo, Pilsung Kang*	2018.11
Comparative Study of Deep Learning-Based Sentiment Classification Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju Seungwan Seo, Czangyeob Kim, Haedong Kim, Kyoung Hyun Mo, Pilsung Kang*	2018.04

Unusual customer response identification and visualization based on text mining and anomaly detection Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju Seungwan Seo, Deokseong Seo, Myeongjun Jang, Jaeyun Jeong, Pilsung Kang*	2018.04
Distance Decomposition for Variable Importance of Distance-based Novelty Detection Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju Seungwan Seo, Myeongjun Jang, Czangyeob Kim, Pilsung Kang*	2018.04
Recurrent Neural Network-based Semantic Variational Autoencoder for Sequence to Sequence Learning Korean Institute of Industrial Engineers (Spring), Hyundai Hotel, Gyeongju Seungwan Seo, Myeongjun Jang, Pilsung Kang*	2018.04
Sentiment classification with word localization based on weakly supervised learning with a convolutional neural network Korean Institute of Industrial Engineers (Fall), KAIST, Daejeon Seungwan Seo, Gichang Lee, Jaeyun Jeong, Czangyeob Kim, Pilsung Kang*	2017.11

Teaching Experiences

Deep Learning Course LG Innotek Full-time Lecture	2022.05
Python Syntax Course LG Innotek Full-time Lecture	2022.05
Python for Deep Learning Course Fastcampus Full-time Lecture	Online course
Python Syntax Course LG Innotek Full-time Lecture	2021.07
Python for Machine Learning and Deep Learning Course SK Hynix Full-time Lecture	2020.11
Python for Machine Learning and Deep Learning Course SK Hynix Full-time Lecture	2020.03
Machine Learning with Python Korea Credit Bureau Full-time Lecture	2019.09
Machine Learning with Python Korea Credit Bureau Full-time Lecture	2019.08
Python for Machine Learning and Deep Learning Course Samsung Electronics Full-time Lecture	2019.07
Python for Machine Learning and Deep Learning Course Samsung Electronics Full-time Lecture	2019.07

Python for Deep Learning for Text Domain

SeoulTech

Seminar

2019.02

Python for Machine Learning and Deep Learning Course

Samsung Electronics

Full-time Lecture

2019.02

Python Syntax Course

Samsung Electronics

Full-time Lecture

2019.01